37 Interpersonal Skills

By David Adler

Topic Highlights

Communication
- *Presentations*
- *Reports*
- *E-Mail Etiquette*
- *Conversations*

People
- *Roles and Responsibilities*
- *Social Styles*
- *Interviews*
- *Negotiations*
- *Conflict Resolution*

Leadership
- *Motivation*
- *Leading Meetings*
- *Leading Teams*
- *Working with Stakeholders*
- *Building Consensus*

37.1 Introduction

This chapter covers the topic of interpersonal skills needed by an automation team member, team leader, or project manager. It covers communicating, writing, interviewing, motivating, negotiating, leading teams, and leading meetings. What an ambitious goal to cover in only a few pages!

No, the author of this topic is not a human resources professional. Rather, I am a 30-year practicing automation professional who has had many leadership opportunities. I also have tuned process control loops, wrote application code for many production facilities, and managed numerous automation projects. I have a passion for the automation discipline and a love for automation professionals. I have found over the years that automation professionals need to 1) know the process and the equipment, 2) have appropriate business processes and project management skills, and 3) have interpersonal skills. A balance of these technical, business, and interpersonal skills are needed on every project or team. All my projects had hardworking and technically competent professionals. The successful projects, however, also had:

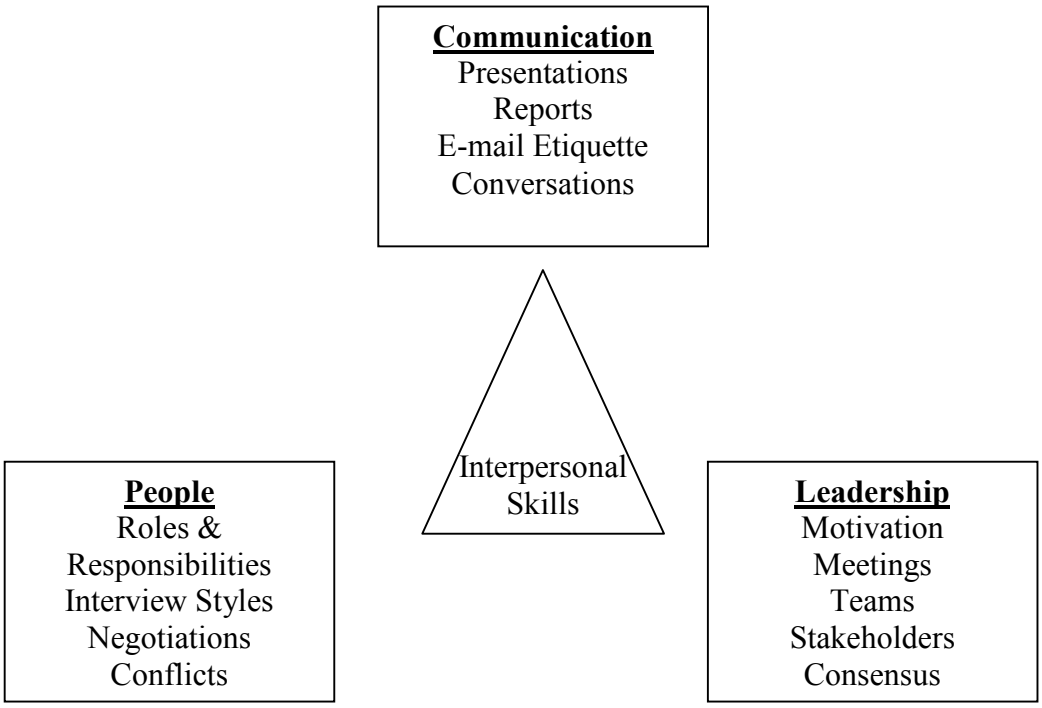


Figure 37-1: Automation Professional Interpersonal Skill Domain

1. a strong purpose and plan,
2. genuine respect and caring among co-workers, and
3. high levels of personal integrity and accountability from all participants.

37.2 Communication

Automation engineers need to communicate their ideas and accomplishments. Yet many technical professionals do not have any confidence in their ability to describe and sell automation projects and technology to their business leaders. With preparation and practice, communication skills can be developed just like any technical skill.

37.2.1 Presentations

What are some general tips for speaking and presenting effectively? (Gruhn 2003)

1. Make sure you have a crisp message. Outline your thoughts ahead of time. Practice.
2. Speak clearly. Manage your pace. Vary your tone. Be enthusiastic. Remember, it is not just what you say but how you say it.
3. Speak with the confidence of an expert, but recognize that the audience may have valuable experiences and insights to share.
4. Look for non-verbals in your listeners (eye contact, posture, facial expression, and hands). If you are losing your audience, ask them a question.

37.2.2 Reports

Take the time to summarize your work in a report. With the pressure of day-to-day assignments increasing all the time, writing a report is lost in the shuffle by the automation engineer when a

project or assignment is completed. However, don't consider your project complete until the final report is finished. What are some rules for successful reports? (Ulman 1972)

1. Consider your audience. Avoid the overuse of process control system jargon. Hopefully, business leaders will want to read your report, and they will want to understand it.
2. Give the reader a clear idea of what it is you are going to talk about at the beginning of the report. A one-page summary and the purpose on the first page of the report is a great way to start.
3. Document the details in the body of the report. Keep it clear and crisp. Don't make it too long. Don't hide your important work under a mountain of minor details. Make liberal use of headings. Make sure to repeat the most important points.
4. Use active verbs. Be positive. Celebrate your and your co-workers' success.
5. Use data in charts and graphs. It is an effective way to reinforce and emphasize key points.
6. Be sure to include the date on every technical report.
7. Be sure to state your conclusions and recommendations.
8. When you are finished with the report, ask yourself if you clearly articulated a crisp theme. Ask a peer to review.

37.2.3 E-mail Etiquette

When you write an e-mail, consider the audience. The ease of sending an electronic message can cause some loss of the quality of your written communications. But the "brain dead" e-mail that was just sent can quickly ripple throughout the whole corporation in a matter of minutes. Take the time to proof it. And if you know the content is controversial, get a trusted peer to read it before you send it out.

37.2.4 Conversations

Automation professionals need to influence others in their daily conversations. Communicating between automation professionals requires you to create a dialogue that is open and honest. It requires speaking and listening skills. The conversation needs to stay focused on what you really want, but it has to be safe for others to talk to you. Be open to critical feedback, and don't take it personally. Speak persuasively and not abrasively. A training associate once told me communicating is just 20% what you say and 80% how you say it. Listen to what is said, understand it, and then respond. Credibility and trust is more a function of good listening than being a good speaker. (Patterson 2002)

37.3 People

Successful teams of automation professionals have the right personnel with solid skills in all the roles needed and a diverse set of social styles. Selecting the right team of automation professionals with all the needed skills is important because people are the most important factor in achieving the automation team's objectives. (Boehm 1981) Automation professionals must also work together on teams and need to learn to negotiate issues and to resolve their conflicts with their peers.

37.3.1 Roles and Responsibilities

As with any major undertaking that involves a group of professionals working together, it is important to define each individual's roles and responsibilities on an automation team. This will ensure each professional knows his area of responsibilities. It will minimize overlap of effort and reduce the possibility of a gap in deliverables. Some examples of roles are automation leader, project manager, automation engineer, testing engineer, and instrument engineer. (Smith 2005)

The *automation leader* is the supervisor of all automation personnel involved. He is accountable for ensuring that design meets operational needs and approves requirement, design, testing plans, and records. He ensures implementation meets corporate standards. He interfaces with site operations and maintenance leaders. He is accountable for testing, acceptance, and start-up.

The *automation project manager* is accountable for meeting the approved scope, schedule, and cost plans. He prepares the project plan. He monitors progress against plan and produces a written report. He coordinates team activities. He interfaces with the overall project leaders and produces project metrics.

The *automation engineer* provides overall technical directions and is responsible for the technical execution of the work. This engineer understands the processing equipment and the control strategy. He defines the automation scope and functional requirements. He designs and implements the automation application. He also interfaces with process engineering and operations.

The *testing engineer* is responsible for the commissioning activities and oversees the documentation process. He ensures the project is delivered to site standards. He drives the completion of automation testing and commissioning activities. He ensures this work is executed in a logical fashion according to plan and cost.

The *instrument engineer* is responsible for the design of the hardware and produces instrument specification, field layout, and design drawings. He ensures the deliverables are listed, understood, and produced on schedule. He coordinates design document review and approval. He oversees procurement and installation. He participates in production start-up.

37.3.2 Social Styles

Social styles can help you understand how to manage interpersonal relationships. One must start by understanding one's own social style. Understanding one's style and others with different styles can help us to relate to each other in more positive ways. Social style does not define our knowledge, experience, or work capacity. The four social styles are driver, expressive, amiable, and analytical. Roughly 25% of a normal population will fall into each of these four styles. (Wilson 1999)

The *driver* gets things done now. He can take independent action and has tight control. To some he can seem overpowering and overly demanding. On an automation project, he will get the project completed on schedule, but he might leave a few dead bodies in his wake.

The *expressive* is extremely creative, has an intuitive understanding of the overall project scope, and can create a lot of excitement and energy. He can be seen as impractical by some and lacking a detail-oriented approach to completing the task. On an automation project, he always has the best overall view of how the whole project fits together, but he often needs help to complete his deliverables.

The *amiable* gets broad acceptance of common objectives and gets everyone to work together in a positive environment. To some, he might be slow to make decisions and seem risk adverse. On an automation project, he is the one who keeps the team working together as a cohesive happy team, but he often is not worried about project costs and schedule.

The *analytical* thinks through problems systematically and does a thorough analysis. He is a deep thinker and solves hard problems. On an automation project, he will solve the most difficult control problem, but he will often get lost in the details and lose sight of the big picture.

There is no "best" social style. Automation professionals can be successful with any social style. It is our flexibility in relating to others with different social styles that promotes success. In fact, I have found it is best to have a balance of all social styles on a project team.

37.3.3 Interviews

The selection of personnel is the most important decision an automation leader makes. No other decision will have a longer lasting impact and be more difficult to undo. Take the extra time to get the

right professional that has technical skills, integrity, and interpersonal skills. It is not uncommon for automation leaders to spend huge blocks of time selecting the process control system vendor and architecture, while spending less time selecting personnel. I have found over the years the right team of automation professionals can make even the worst vendors' hardware and software systems work successfully. Let me select the team of automation professionals and let anyone pick the distributed control system (DCS) or programmable logic controller (PLC); the team will make it work. A key skill in selecting the right personnel is the interview process. (Byham 1979)

Before you start interviewing, make sure you have defined the needed roles and responsibilities for the automation team. Write a job description. Acquire the needed training and guidance to know legally what questions are off limits. Develop the questions you will ask before the interview. Review the applicant's resume. Three key areas of focus for these questions are 1) leadership and problem solving skills, 2) team work and interpersonal skills, and 3) technical proficiency.

At the start of the interview, build rapport with the candidate, as it puts him or her at ease. Describe the position and the selection process. Take written notes. Manage the pace. Keep the candidate focused or draw him or her out, as appropriate. Avoid making a quick decision in the first few minutes of an interview. Focus on all three specific areas. Don't let the candidate talk about what "we" did, but probe to find out in particular what "I" did.

37.3.4 Negotiations

Many automation professionals need to negotiate with their peers, customers, and vendors. With a systemic approach and preparation, an automation professional can increase his chances of getting the outcome he wants. Take the time to prepare when negotiating. Some negotiating techniques are: take it or leave it, silence, surprise, good cop/bad cop, and deadlock. What are the keys to successful negotiation? (Benedict 2005)

1. Know your leverage. Aim high. Know what it is you want.
2. The first few minutes are important, so watch your words. Don't use weak words such as "we like" or "we prefer" but rather use "I need" or "It's essential." Take control of the meeting at the beginning of negotiations. Don't be afraid to change the agenda to your advantage. Stay assertive, confident, and non-manipulative.
3. Trade rather then make concessions. Give each concession its highest psychological value. Look for trades of high value to them, but of low value to you.
4. Negotiate with honesty and integrity.

37.3.5 Conflict Resolution

Conflict is a normal part of automation projects. The need to balance schedule, cost, and scope, while maintaining high quality instrumentation design, application development, and process control capability, creates tension. This tension can help engineers to think harder and create even better solutions. But if two strongly opinionated automation professionals choose different paths for the technical solution, and then make it personal, it can result in conflict. Then there is a need to manage the disagreement.

Common responses to conflict are avoid, minimize, attack, compromise, and problem-solve. What do you do if you find yourself in a serious conflict? (Davis 1992)

1. Listen. Put yourself in the other person's situation. Look at the other person's point of view. Don't lecture on why you are right. Attack the problem and not the person.
2. Understand the underlying problem. When you think a topic has been discussed, summarize and ask the other person if all the issues are on the table.

3. Allow the other person to vent. If it gets out of hand, stop the discussion and try to solve the conflict another day.

4. Find areas of mutual agreement. Clearly state the area of agreement and then try to resolve any remaining issues.

If this is a recurring issue, or the parties involved can't resolve the conflicts, the leader shouldn't hesitate to get involved. The team leader needs to help the automation professionals focus on facts and not on personalities. Make sure each is listening, and find common ground. Set up a plan to resolve and follow-up. Don't let the conflict fester and impact the whole team. If necessary, get help from a human resources professional.

37.4 Leadership

Automation professionals are often asked to take a leadership role in projects, daily assignments, and meetings. The leader of a successful team must have: 1) good communication skills, 2) good people skills, and 3) good decision-making skills. (Scholtes 2003) Leaders need to serve the other team members. (Block 1996) They need to see the value of influencing individuals as an opportunity for each team member to develop his or her unique gifts. The leader motivates and builds consensus to achieve the common objective. She or he needs to inspire the team members and work with the team's many stakeholders.

37.4.1 Motivation

Individuals must be able to find their own motivation to perform. As an automation leader or project manager, you can provide an environment in which personal motivation will flourish. The top cause of loss of motivation is lack of control over one's job. (Connors 1994) An automation leader needs to create an environment of accountability and help team members understand they have responsibility over their work. The leader needs to ensure work/life balance and lead by example. The automation team leader needs to make coaching team members a priority. How does a team leader coach? (Davis 1992)

1. Develop a partnership between leader and automation professional. Trust each other. This first step is often the hardest. Ask each other for feedback. Mutually develop a few critical objectives and priorities.

2. Promote persistence and self-discipline. The leader focuses on results and not just activity. The leader holds the team member accountable to agreed commitments and expects results. He or she celebrates the successes and encourages in a positive way the need to improve.

3. The leader removes barriers, eliminating unnecessary permissions and approvals. She or he gives the support needed.

4. The leader encourages lifetime learning. New skills allow engineers to perform at the highest level.

A good leader also recognizes when it is not appropriate to coach—for example, when the employee is at capacity and any more feedback will overwhelm; the setting is public, such as a group meeting; the leader is not prepared, pressed for time, or preoccupied with other problems; and when the feedback is too general or dated to be of value. Feedback needs to be timely and constructive.

37.4.2 Leading Meetings

Many automation professionals dislike meetings. They see them as an obstacle or time away from getting the "real" engineering work done. But the need to sell your ideas often are best done in a meeting. The needed group consensus can often be generated by the group discussions and decisions in a

good team meeting. By following a few simple rules of pre-work and meeting management, it is possible to achieve the desired outcome from a meeting.

Take the time to plan the meeting. Define the meeting's purpose. Make sure there is business value in having the participants come together. Develop an agenda and include a list of topics to discuss, and expected results. Define the length. Don't schedule the meeting for longer than is needed. Prepare presentation and handouts. Make the necessary logistical arrangements for audio, video, and/or teleconferencing. How do you lead the meeting? (Streibel 2003)

1. Start the meeting on time. Establish the meeting rules.
2. Assign a facilitator to manage time, keep the group on agenda, and drive the meeting to the desired outcomes.
3. Manage the tone and tempo. Make sure all participate. Don't allow just a few to dominate the discussion. Manage the participants' need to digress into side issues. Have fun.
4. Drive the team to the expected results. Make a decision. Finish on time.
5. Take minutes and publish.

The best way to improve meetings is to make sure that every participant attends with a sense of the meeting's purpose. This is accomplished by publishing an agenda before the meeting to give participants ample time to prepare. Another way to improve meetings is to ask the meeting participants for feedback. Take five minutes at the end of the meeting and give each participant a chance to comment on what went well and what needs to be improved. Finally, if the meeting was worth having, make sure it generates action. Publish the minutes and action items. Follow-up and ensure action item completion.

37.4.3 Leading Teams

Fostering teamwork requires focus by the automation team leader. In today's complex automation environment, delivering projects, and resolving problems would be impossible without teams. No individual automation professional can know all there is to know about instrumentation, system applications, and automation. Teams go through normal dynamics that can be categorized as forming, storming, norming, and then performing. (Levi 2001)

The Team Forms. When a team forms, it can generate a lot of excitement and positive energy. But some team members can also be concerned and uncertain about the new activity. The team builds its relationships. The leader must clearly and concisely articulate the team's purpose and priorities.

The Team Storms. Before a team can get to the highest level of performance, it must learn to work together. This requires testing roles, authority, and boundaries. Team members learn to work through differences of opinions. The team leader encourages equal participation, and the team grows from the tension as ideas are discussed.

The Team Norms. A team has learned to work through differences of opinions and to solve common problems. They accept the team's norms and each other's roles. They are able to give each other feedback. Each team member feels empowered, works hard, and is able to solve problems. The project is now making progress. The team leader is able to share significant responsibilities.

The Team Performs. The individual team members now cooperate and perform as a cohesive team. The team delivers at a higher level than the sum of each individual's efforts. Each team member supports and complements the other team members to achieve the highest level of performance. Individuals not only take pride in their work, they also take satisfaction in the overall contribution of the whole team. The leader celebrates the success of the team.

37.4.4 Working with Stakeholders

Automation professionals may have many stakeholders. These stakeholders might include the leadership of operations, process engineering, plant engineering, shops, information technology, technical services, quality control, and numerous manufacturing business leaders. To work effectively with stakeholders on a difficult issue or project requires: 1) understanding their problems, 2) delivering good workable solutions, and 3) anticipating future needs. (Block 2000)

Understand their problems. Take the time to meet directly with each stakeholder. Find out each stakeholder's expectations, and state clearly your desire for a balanced relationship. Be willing to say "no" to unreasonable stakeholder requests. Develop a service-level agreement with specific deadlines and get the stakeholders to sign off. Remember the purpose of all this planning is to generate action.

Deliver solutions. Implementation is more difficult than planning. Make sure you get your hands dirty in the details of the problem. Solicit feedback throughout from stakeholders. Understand that stakeholder criticism and resistance is not directed at you personally. Persevere in solving any issues. Train operators and teach them to solve problems. At the end of a project, take the time to celebrate your success. And for those few things that did not go well, learn from it and forgive.

Anticipate future needs. Discuss with stakeholders your strengths and weaknesses. Don't assume that the stakeholders' understanding is the same as yours. Be committed to never ending improvement. Grow new skills and see learning as an adventure. Engage with the stakeholders in long-range planning.

The automation professional must try to influence stakeholders—often without any direct power to make changes or implement programs. The goal is to develop a collaborative relationship with stakeholders, to solve their problems so they stay solved, and to ensure attention is given to both technical problems and interpersonal relationships. Trust and respect from stakeholders are gained by completing each requirement in a quality manner, on schedule, and within cost.

37.4.5 Building Consensus

An automation organization needs to have focus and alignment to achieve results. A common purpose is the best way to build consensus among automation team members. Without a common purpose, the team's effort will be random, and it will take heroic efforts and good luck to achieve results. The four steps are: 1) understand current givens, 2) define the future state, 3) define changes needed, and 4) plan and implement.

Understand the current givens. Understand your operation's purpose and trends in automation technology. Benchmark other automation groups. Define your team's strengths and weaknesses. This step defines clearly your current state.

Define the future state. The automation team develops its purpose. What does this automation team want to accomplish, and how will we achieve it? Define the new investments needed. This step develops a high level view of the future that automation will bring to support production operations or to implement a project.

Define changes needed. How will we align organizational structure, leadership, people, management, work process, and culture to achieve? Define the business value this will bring to our customers. This step defines the new capabilities needed to move to this new future state.

Plan and implement. The final step is to develop and execute an operational plan. Develop specific scope, schedules, and costs to achieve. Develop an action plan and track progress.

Automation leaders must have a clear purpose and articulate it. They must build consensus in their organizations and inspire team members and stakeholders with their enthusiasm. Building consensus depends on the support of your stakeholders. Equally important is getting the active engagement and buy-in from every team member. Explain the rationale. Ask each team member or stakeholder for their view of the future and how automation fits into the success of operations. And, finally, be pre-

pared to give this crisp message many times. A wise production manager once told me, if you wish to achieve change and have everyone buy-in, you need to be prepared to give your pep talk 37 different times.

37.5 Conclusion

People are the most important factor in an automation team achieving its objectives. Develop your interpersonal skills just like you develop your technical skills. Encourage and support other team members to develop and maximize their communication, people, and leadership skills.

If you find your interest has been sparked by this topic, I hope you will read one of the many excellent books on interpersonal skills referenced below and in this chapter.

37.6 References

1. Benedict, Robert. "Real World Negotiating." (Seminar) Benedict Negotiating Seminars, Inc. Valrico, Florida. Web site: www.backdoorselling.com

2. Block, Peter. *Flawless Consulting: A Guide to Getting Your Expertise Used.* Jossey-Bass Pfeiffer, 2000.

3. Block, Peter. *Stewardship: Choosing Service Over Self-Interest.* Berrett-Koehler Publishers, 1996.

4. Boehm, Barry W. *Software Engineering Economics.* Prentice Hall, 1981.

5. Byham, William C. *Targeted Selection®: A Behavioral Approach to Improved Hiring Decisions.* (Monograph) Development Dimensions International, 1979 (revised 2004). Website: www.ddiworld.com

6. Connors, Roger, Tom Smith, and Craig Hickman. *The Oz Principle: Getting Results Through Individual and Organizational Accountability.* Prentice Hall, 1994.

7. Davis, Brian, et al. *Successful Managers Handbook: Development Suggestions for Today's Managers.* Personnel Decisions International, 1992.

8. Gruhn, Paul. *Sell More Through Effective Technical Presentations—A pocket guide.* ISA, 2003.

9. Levi, Daniel. *Group Dynamics for Teams.* Sage Publications, 2001.

10. Patterson, Kerry, et al. *Crucial Conversations: Tools for Talking When Stakes are High.* McGraw-Hill, 2002.

11. Scholtes, Peter E., Brian L. Joiner, and Barbara J. Streibel. *The Team Handbook*. Third Edition. Oriel, 2003.

12. Smith, Graham. *Managing Software Contractors*. Brillig Systems, 2005. Website: www.brilligsys.com

13. Streibel, Barbara J. *The Manager's Guide to Effective Meetings.* McGraw-Hill, 2003.

14. Ulman, Joseph N. and Jay R. Gould. *Technical Reporting.* Third Edition. Holt, Rinehart, and Winston, 1972.

15. Wilson Learning Corporation. *Social Styles as a Global Phenomenon*. (White Paper) Wilson Learning Corporation, 1999. Web site: www.wilsonlearning.com (retrieved 06/21/2005)

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